

Exercise 6: Multi-Layer Perceptrons - Forward Pass

Task 1:

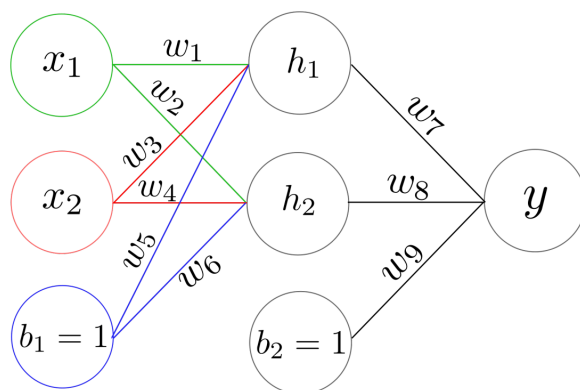
Explain the role and effects of a bias neuron in artificial neural networks.

Task 2:

The value of a neuron is a function of the input and the associated weights. Accordingly, how can the output of a multilayer neural network be expressed mathematically? Using an example of your choice, describe how the output of a multilayer neural network is determined, i.e. how inference works.

Task 3

When training multi-layer perceptrons (aka feedforward networks) the first steps are performing a forward pass and computing the error of the prediction. Calculate the forward pass for the following MLP and input. Thus, calculate the values of the two hidden neurons and the result y . Additionally, calculate the error of the prediction.



- Weights: $w_1 = 1.2, w_2 = 0.7, w_3 = 0.8, w_4 = -0.6, w_5 = -0.9, w_6 = -0.2, w_7 = -0.5, w_8 = 0.5, w_9 = 0.2$
- The hidden neurons, as well as the output neuron use the threshold activation function:
 $f(x) = 1, \text{ für } x \geq 0,$
 $f(x) = 0, \text{ für } x < 0$
- Input: $x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, y = 1$
- Error function: $\mathcal{L} = \frac{1}{2}(o - y)^2$ with o being the expected value and y being the result of the forward pass.

Task 4

Implement the network above using only native Python and numpy, so that it can perform the forward pass. Use the same values for its parameters.

Test it using the input $x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and compare the result and error to your manual calculations.

Note: Instead of just doing the calculations manually in the code, try to implement a class-based implementation for the MLP, that allows for the flexible construction of the MLP out of Layer and Neuron objects.